

ORIGINAL ARTICLE

Mapping the HLA ligandome landscape of acute myeloid leukemia: a targeted approach toward peptide-based immunotherapy

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Identification of physiologically relevant peptide vaccine targets calls for the direct analysis of the entirety of naturally presented human leukocyte antigen (HLA) ligands, termed the HLA ligandome. In this study, we implemented this direct approach using immunoprecipitation and mass spectrometry to define acute myeloid leukemia (AML)-associated peptide vaccine targets. Mapping the HLA class I ligandomes of 15 AML patients and 35 healthy controls, more than 25 000 different naturally presented HLA ligands were identified. Target prioritization based on AML exclusivity and high presentation frequency in the AML cohort identified a panel of 132 LiTAAs (ligandome-derived tumor-associated antigens), and 341 corresponding HLA ligands (LiTAPs (ligandome-derived tumor-associated peptides)) represented subset independently in >20% of AML patients. Functional characterization of LiTAPs by interferon- γ ELISPOT (Enzyme-Linked ImmunoSpot) and intracellular cytokine staining confirmed AML-specific CD8⁺ T-cell recognition. Of note, our platform identified HLA ligands representing several established AML-associated antigens (e.g. NPM1, MAGE1, PRTN3, MPO, WT1), but found 80% of them to be also represented in healthy control samples. Mapping of HLA class II ligandomes provided additional CD4⁺ T-cell epitopes and potentially synergistic embedded HLA ligands, allowing for complementation of a multi-peptide vaccine for the immunotherapy of AML.

Leukemia advance online publication, 29 August 2014; doi:10.1038/leu.2014.233

INTRODUCTION

A growing body of evidence demonstrates that peptide-based cancer immunotherapy can induce specific immune responses and impact clinical outcome.^{1–5} Recent data even indicate that induction of specific T-cell responses by tumor-specific multi-peptide vaccination prolongs overall survival in cancer patients.⁶ In acute myeloid leukemia (AML), the persistence of residual leukemic cells (i.e., minimal residual disease) leads to high relapse rates after standard polychemotherapy and also after allogeneic stem cell transplantation, which results in poor overall survival.^{7–10} Immunogenicity of AML, as revealed in graft-versus-leukemia effects, as well as in favorable immune effector-to-target cell ratios present in the minimal residual disease setting, suggests that AML might be effectively targeted by multi-peptide immunotherapy.^{11,12} Human leukocyte antigen (HLA) surface expression on AML blasts, a prerequisite for T-cell recognition, has been controversially discussed in recent years.^{13,14} Furthermore, in contrast to solid tumors, few leukemia-associated antigens, including some naturally presented phosphopeptides, have been described so far.^{15–17} Some of these antigens including WT1,^{18,19} PR1^{5,20} and RHAMM^{3,21} have found their way into clinical phase I/II trials, showing promising results in terms of *in vivo* immunogenicity as well as clinical responses in single patients.^{1,4,22–24} However, broad applicability and induction of clinical responses

have not been achieved so far. In recent years, we demonstrated that there is only a poor correlation of gene expression and HLA-restricted presentation of corresponding gene products,^{25,26} calling for direct methods of target identification. To characterize differences between AML and normal PBMCs and BMNCs (peripheral blood mononuclear cells and bone marrow mononuclear cells) and to identify physiologically relevant tumor-associated peptides, we implemented the approach of direct elution and identification of naturally presented HLA ligands from primary samples by immunoprecipitation and liquid chromatographic tandem mass spectrometry (LC-MS/MS). Implementation of nanoscale liquid chromatography (nano-UHPLC) and modern hybrid mass spectrometry resulted in highly sensitive detection of peptides and vastly improved numbers of HLA ligand identifications as compared with previous studies.^{27,28} Out of the vast array of potential targets, we selected the most relevant and broadly applicable candidates by devising a novel target identification platform focused on identifying shared tumor-associated self-antigens. In this study, we based target prioritization primarily on AML exclusivity, followed by a high presentation frequency of the antigen in the HLA ligandomes of primary AML patient samples. This novel platform provided us with an array of >1700 AML-exclusive antigens represented by more than 2400 HLA class I and II ligands. A subset of 168 LiTAAs (ligandome-

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Received 23 April 2014; revised 23 July 2014; accepted 24 July 2014; accepted article preview online 5 August 2014

derived tumor-associated antigens) was defined based on representation in >20% of AML ligandomes. Corresponding HLA ligands, termed LiTAPs (ligandome-derived tumor-associated peptides), were functionally further characterized providing evidence for AML-specific immune recognition.

MATERIALS AND METHODS

Patients and blood samples

For ligandome analysis, PBMCs from AML patients at the time of diagnosis or at relapse before therapy (>80% of AML blast count in blood), as well as PBMCs and BMNCs of healthy donors, were isolated by density gradient centrifugation. Informed consent was obtained in accordance with the Declaration of Helsinki protocol. The study was performed according to the guidelines of the local ethics committee. Patient characteristics are provided in Table 1. HLA typing was carried out by the Department of Hematology and Oncology, Tübingen, Germany. Samples were stored at –80 °C until further use.

Quantification of HLA surface expression

To allow for comparison with healthy monocytes, quantification of HLA surface expression was performed in additional patient samples containing CD15⁺ AML blasts and at least 5% normal CD15⁺ monocytes as defined by immunophenotyping (Supplementary Table S1). HLA surface expression was analyzed using the QIFIKIT quantitative flow cytometric assay kit (Dako, Carpinteria, CA, USA) according to the manufacturer's instructions. In brief, triplicates of each sample were stained with the pan-HLA class I-specific monoclonal antibody (mAb) W6/32, HLA-DR-specific mAb L243 (both produced in-house) or IgG isotype control (BioLegend, San Diego, CA, USA), respectively. Secondary staining with FITC-conjugated rabbit-anti-mouse F(ab')₂ fragments (Dako) was subsequently carried out on PBMCs and BMNCs using QIFIKIT quantification beads (Dako). Surface marker staining was carried out with directly labeled CD34 (BD Bioscience, Franklin Lakes, NJ, USA), CD15 (BD Bioscience), CD45 (BD Bioscience) and CD38 (BD Bioscience). 7-Aminoactinomycin D (BioLegend) was added as viability marker immediately before flow cytometric analysis on an LSR Fortessa cell analyzer (BD Bioscience).

Isolation of HLA ligands from primary samples

HLA class I and II molecules were isolated using standard immunoaffinity purification as described previously.^{29,30} In brief, snap-frozen cell pellets were lysed in 10 mM CHAPS/PBS (3-[3-cholamidopropyl] dimethylammonio]-1-propanesulfonate/phosphate-buffered saline) (AppliChem, St Louis,

MO, USA/Gibco, Carlsbad, CA, USA) containing 1x protease inhibitor (Complete; Roche, Basel, Switzerland). HLA molecules were single-step purified using the pan-HLA class I-specific mAb W6/32 and the pan-HLA class II-specific mAb Tü39, respectively, covalently linked to CNBr-activated sepharose (GE Healthcare, Chalfont St Giles, UK). HLA-peptide complexes were eluted by repeated addition of 0.2% trifluoroacetic acid (Merck, Whitehouse Station, NJ, USA). Elution fractions E₁–E₈ were pooled and free HLA ligands were isolated by ultrafiltration using centrifugal filter units (Amicon; Millipore, Billerica, MA, USA). HLA ligands were extracted and desalted from the filtrate using ZipTip C₁₈ pipette tips (Millipore). Extracted peptides were eluted in 35 µl of 80% acetonitrile (Merck)/0.2% trifluoroacetic acid, centrifuged to complete dryness and resuspended in 25 µl of 1% acetonitrile/0.05% trifluoroacetic acid. Samples were stored at –20 °C until analysis by LC-MS/MS.

Analysis of HLA ligands by LC-MS/MS

Peptide samples were separated by reversed-phase liquid chromatography (nano-UHPLC, UltiMate 3000 RSLCnano; Thermo Fisher, Waltham, MA, USA) and subsequently analyzed in an online coupled LTQ Orbitrap XL hybrid mass spectrometer (Thermo Fisher). Samples were analyzed in five technical replicates. Sample volumes of 5 µl (sample shares of 20%) were injected onto a 75 µm × 2 cm trapping column (Acclaim PepMap RSLC; Thermo Fisher) at 4 µl/min for 5.75 min. Peptide separation was subsequently performed at 50 °C and a flow rate of 175 nl/min on a 50 µm × 50 cm separation column (Acclaim PepMap RSLC; Thermo Fisher) applying a gradient ranging from 2.4 to 32.0% of acetonitrile over the course of 140 min. Eluting peptides were ionized by nanospray ionization and analyzed in the mass spectrometer implementing a top five CID (collision-induced dissociation) method generating fragment spectra for the five most abundant precursor ions in the survey scans. Resolution was set to 60 000. For HLA class I ligands, the mass range was limited to 400–650 *m/z* with charge states 2 and 3 permitted for fragmentation. For HLA class II, a mass range of 300–1500 *m/z* was analyzed with charge states ≥ 2 allowed for fragmentation.

Database search and spectral annotation

For data processing, the software Proteome Discoverer (v.1.3; Thermo Fisher) was used to integrate the search results of the Mascot search engine (Mascot 2.2.04; Matrix Science, London, UK) against the human proteome as comprised in the Swiss-Prot database (<http://www.uniprot.org>, release: 27 September 2013; 20 279 reviewed protein sequences contained). Of note, this database assisted search only allows for the identification of non-mutant human self-peptides. The search combined

Table 1. Patient characteristics

UPN	WHO	FAB	Molecular genetic	Karyotype	HLA typing	WBC (10 ³ /µl) (% blast)	Cell count (10 ⁹)
1	MDS/MPS related	M2	NM	46, XX	A*02, A*11, B*35, B*44	11.0 (80)	0.9
2	Therapy related	M5	NM	Complex	A*02, A*31, B*27, B*44	140.9 (94)	11.0
3	MDS/MPS related	M2	JAK2	Complex	A*23:01, A*66:01, B*49:01	29.2 (83)	0.5
4	Recurrent genetic abnormalities	M1	FLT3-ITD, NPM1_A	46, XX	A*03:01, B*39:01, B*51:01	248.1 (80)	1.0
5	Recurrent genetic abnormalities	M1	FLT3-ITD	46, XX	A*02:01, B*07:02, B*40:01	83.0 (90)	0.4
6	Recurrent genetic abnormalities	M5	NPM1_A	46, XX	A*02, A*03, B*44:25, B*52:15	50.0 (84)	1.6
7	Recurrent genetic abnormalities	M5	FLT3-ITD, NPM1_A	46, XX	A*02:01, A*03:01, B*38:01, B*44:02	38.0 (84)	2.8
8	MDS/MPS related	M1	NM	Complex	A*03:01, A*26:01, B*35:01, B*38:01	47.5 (90)	1.7
9	Recurrent genetic abnormalities	M2	FLT3-ITD	46, XY	A*02:01, A*66:01, B*40:01, B*15:01	187.4 (80)	8.4
10	Recurrent genetic abnormalities	M4	CBFB/MYH11	47, XY, inv(16)(p13.1q22)	A*02:01, B*51:01, B*39:01	27.0 (80)	0.4
11	Recurrent genetic abnormalities	M5	FLT3-ITD, NPM1_A	46, XY	A*02:01, A*23:01, B*44:02, B*49:01	163.8 (90)	0.4
12	Recurrent genetic abnormalities	M2	FLT3-ITD	46, XY	A*02:01, B*13:02, B*51:01	24.5 (80)	0.2
13	MDS/MPS related	M1	NM	45, XY, –7, +13, –21	A*02:01, A*03:01, B*18:01	39.2 (84)	2.6
14	Recurrent genetic abnormalities	M2	FLT3-ITD, NPM1_A	46, XX	A*02:01, A*26:01, B*27:05, B*51:01	150.0 (96)	4.9
15	MDS/MPS related	M1	NM	Complex	A*29:01, A*32:01, B*40:01, B*44:03	222.6 (93)	3.5
16	Recurrent genetic abnormalities	M5	FLT3-ITD	46, XY	A*02, A*24, B*44, B*50	250.0 (92)	19.0
17	MDS/MPS related	M2	NM	45, XY, –7	A*26, A*32, B*15, B*38	260.0 (80)	7.6

Abbreviations: AML, acute myeloid leukemia; FAB, French-American-British; FLT3-ITD, FMS-like tyrosine kinase 3-internal tandem duplication; HLA, human leukocyte antigen; MDS/MPS related, AML with myelodysplasia-related/myeloproliferative-related changes; NM, no mutation; recurrent genetic abnormalities, AML with recurrent genetic abnormalities; therapy related, therapy-related myeloid neoplasms; WBC, white blood cell count; WHO, World Health Organization; UPN, uniform patient number. Patients included in HLA class I and II ligandome analysis. WHO classification and complex karyotype (≥3 unrelated chromosomal abnormalities) analogous to Swerdlow *et al.*,⁵⁵ WHO Classification of Tumors of Hematopoietic and Lymphoid Tissues. Patients 1, 8 and 11 were included in the study at the time of relapse. Cell count specifies the number of isolated PBMCs for HLA ligand isolation. For patients 2 and 10, only HLA class II ligandome profiling could be successfully executed.

data of technical replicates and was not restricted by enzymatic specificity. Precursor mass tolerance was set to 5 p.p.m., and fragment mass tolerance to 0.5 Da. Oxidized methionine was allowed as a dynamic modification. False discovery rates were determined by the Percolator algorithm³¹ based on processing against a decoy database consisting of the inverted target database. False discovery rate was limited to $q < 0.05$ (5% false discovery rate). Peptide-spectrum matches (PSMs) with $q < 0.05$ were filtered according to additional, orthogonal parameters to ensure spectral quality and validity. Mascot scores were filtered to ≥ 20 . For HLA class I, peptide lengths were limited to 8–12 amino acids in length. For HLA class II, peptides were limited to 12–25 amino acids in length. Protein grouping was disabled, allowing for multiple annotations of peptides (e.g. conserved sequences mapping into multiple proteins). As a final step of quality control thresholds of ≥ 500 (HLA class I, AML and PBMC), ≥ 200 (HLA class I, BMNC) and ≥ 100 (HLA class II, all samples), unique ligand identifications per sample were applied. HLA annotation was performed by *in silico* motif mapping using SYFPEITHI (<http://www.syfpeithi.de>) or an extended in-house database. Experimental validation of the predicted HLA restriction was obtained by peptide synthesis and functional characterization for a small subset of ligands.

Peptide synthesis

The automated peptide synthesizer EPS221 (Abimed, Langenfeld, Germany) was used to synthesize peptides using the 9-fluorenylmethyl-oxycarbonyl/tert-butyl (Fmoc/tBu) strategy as described.³² Synthetic peptides were used for the validation of mass spectrometric identifications as well as for functional experiments.

Quantitative analysis of *WT1* mRNA expression

Relative *WT1* mRNA expression levels were analyzed by RT-qPCR and normalized to *ABL* mRNA levels. The analysis was performed by MLL Laboratories (Münchner Leukämie Labor GmbH, München, Germany).

Amplification of peptide-specific T cells

PBMCs from AML patients and healthy volunteers were cultured in RPMI1640 medium (Gibco) supplemented with 10% pooled human serum (produced in-house), 100 mM β -mercaptoethanol (Roth, Karlsruhe, Germany) and 1% penicillin/streptomycin (GE Healthcare, Fairfield, CT, USA). For CD8⁺ T-cell stimulation, PBMCs were thawed and pulsed with 1 μ g/ml per peptide. Peptide-pulsed PBMCs ($5\text{--}6 \times 10^5$ cells per ml) were cultured at 37 °C and 5% CO₂ for 12 days. On days 0 and 1, 5 ng/ml IL-4 (R&D Systems, Minneapolis, MN, USA) and 5 ng/ml IL-7 (Promokine, Heidelberg, Germany) were added to the culture medium. On days 3, 5, 7 and 9, 2 ng/ml IL-2 (R&D Systems) were added to the culture medium. Peptide-stimulated PBMCs were functionally characterized by ELISPOT (Enzyme-Linked ImmunoSpot) assays on day 12 and by intracellular cytokine staining on day 13, respectively. For CD4⁺ T-cell stimulation, culture was performed as described for CD8⁺ T cells with two modifications: pulsing was carried out with 10 μ g/ml of HLA class II peptide and no IL-4 and IL-7 was added.

IFN- γ ELISPOT assay

Interferon- γ (IFN- γ) ELISPOT assays were carried out as described previously.³³ In brief, 96-well nitrocellulose plates (Millipore) were coated with 1 mg/ml IFN- γ mAb (Mabtech, Cincinnati, OH, USA) and incubated overnight at 4 °C. Plates were blocked with 10% pooled human serum for 2 h at 37 °C. A total of 5×10^5 cells per well of prestimulated PBMCs were pulsed with 1 μ g/ml (HLA class I) or 2.5 μ g/ml (HLA class II) peptide and incubated for 24–26 h. Readout was performed according to the manufacturer's instructions. Spots were counted using an ImmunoSpot S5 analyzer (CTL, Shaker Heights, OH, USA). T-cell responses were considered to be positive when >15 spots per well were counted and the mean spot count per well was at least threefold higher than the mean number of spots in the negative control wells (according to the cancer immunoguiding program guidelines³⁴).

Intracellular IFN- γ and TNF- α staining

The frequency and functionality of peptide-specific CD8⁺ T cells was analyzed by intracellular IFN- γ and TNF- α staining as described previously.^{27,33} PBMCs were pulsed with 1 μ g/ml of individual peptide and incubated in the presence of 10 μ g/ml brefeldin A (Sigma, St Louis,

MO, USA) and 10 μ g/ml GolgiStop (BD Bioscience) for 6–8 h. Cells were labeled using Cytofix/Cytoperm (BD Bioscience), CD8-PECy7 (Beckman Coulter, Fullerton, CA, USA), CD4-APC (BD Bioscience), TNF- α -PE (Beckman Coulter) and IFN- γ -FITC (BD Bioscience). Samples were analyzed on a FACS Canto II (BD, Franklin Lakes, NJ, USA).

Software and statistical analysis

Flow cytometric data analysis was performed using FlowJo 7.2 (Tree Star, Ashland, OR, USA). GraphPad Prism 6.0 software (GraphPad Software Inc., La Jolla, CA, USA) was used for statistical analysis implementing the nonparametric Wilcoxon's matched-pairs signed-rank test and Spearman's rank correlation coefficient, where appropriate. Overlap analysis was performed using the BioVenn tool (<http://www.cmbi.ru.nl/cdd/biovenn>).

RESULTS

Primary AML samples display no loss or downregulation of HLA expression compared with autologous benign leukocytes

As a first step, we aimed to determine the HLA expression levels on AML blasts compared with corresponding benign leukocytes. To this end, HLA surface levels were quantified by flow cytometry in a panel of five patients with CD15⁺ AML (Supplementary Table S1) and five healthy BMNC donors. AML blasts were gated as CD34⁺, CD45^{med} viable cells, and their HLA expression was compared with autologous CD15⁺ normal granulocytes and monocytes. HLA levels were found to be heterogeneous with total HLA class I molecule counts ranging from 45 189 to 261 647 molecules per cell on AML blasts and 75 344 to 239 496 molecules per cell on CD15⁺ cells. Patient individual analysis of HLA surface expression in triplicates is shown in Figure 1a. HLA-DR expression ranged from 1476 to 45 150 molecules per cell on AML blasts and 0 to 3252 on CD15⁺ cells (Figure 1b). Comparison of mean HLA class I and II expression on AML blasts and autologous normal monocytes revealed no significant difference (Figures 1c and d). For reference, HLA surface molecule counts on hematopoietic progenitor cells (CD34⁺CD38[−]) of five healthy BMNC donors were analyzed. No significant difference between mean HLA class I count on normal hematopoietic progenitor cell ($248\,587 \pm 35\,351$ molecules per cell) and AML blasts ($116\,445 \pm 37\,855$ molecules per cell; Figure 1e) was detected. Mean HLA class II count on normal hematopoietic progenitor cell ($38\,373 \pm 5159$ molecules per cell) showed no significantly elevated level compared with AML blasts ($17\,103 \pm 7604$ molecules per cell; Figure 1f).

LC-MS/MS identifies a vast array of naturally presented HLA class I and II ligands

Mapping the HLA class I ligandomes of 15 AML patients (Table 1), we were able to identify a total of 13 238 different peptides representing 6104 source proteins. The number of identified unique peptides per patient ranged from 563 to 2733 (mean 1299 peptides; Figure 2). The reproducibility and variation of the technical LC-MS/MS replicates is shown in Supplementary Figure S1. In the healthy cohort (30 PBMC donors, 5 BMNC donors), a total of 17 940 unique peptides were identified (17 322 peptides/7207 source proteins on PBMCs; 1738 peptides/1384 source proteins on BMNCs; Supplementary Table S2). Analysis of HLA class II ligandomes in 12 AML patients yielded a total of 2816 unique peptides (range 104–753 peptides/patient, mean 332 peptides; Figure 2) representing 885 source proteins. The HLA class II healthy control cohort (13 PBMCs, 2 BMNC donors) yielded 2202 different peptides (2046 peptides/756 source proteins on PBMCs; 317 peptides/164 source proteins on BMNCs; Supplementary Table S2). No correlation of analyzed cell numbers and the number of peptide identifications was found either for HLA class I (Spearman's $r=0.27$, $P=0.33$) or for HLA class II ($r=0.31$, $P=0.33$).

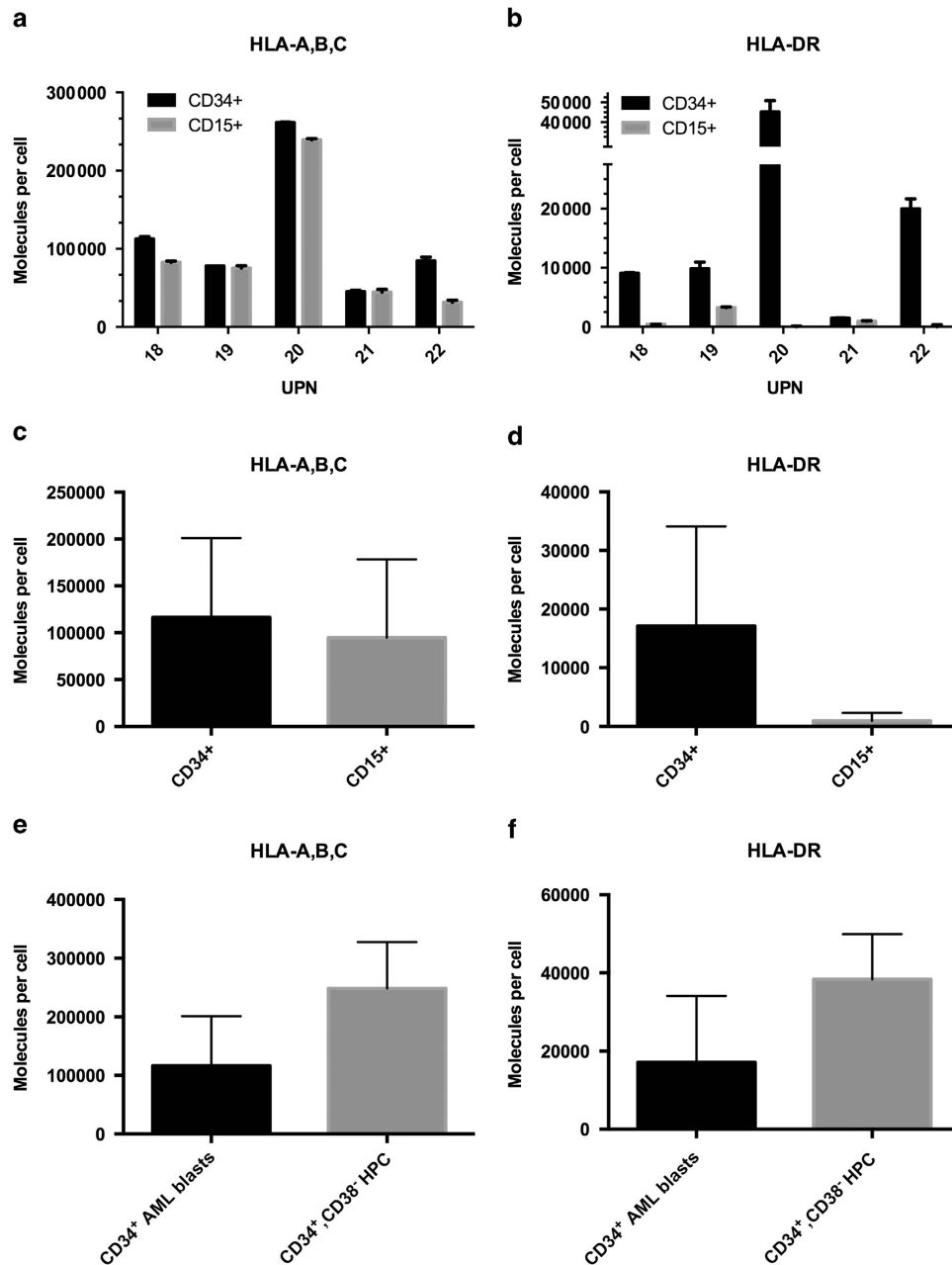


Figure 1. HLA surface expression of primary AML samples and healthy donor hematopoietic progenitor cell. Quantification was performed *ex vivo* using QIFIKIT (Dako). (a) HLA class I (W6/32 mAb) expression of CD34⁺ AML blasts compared with autologous CD15⁺ normal monocytes. (b) HLA-DR (L243 mAb) expression of CD34⁺ AML blasts compared with autologous CD15⁺ normal monocytes. (c) Mean HLA class I and (d) HLA class II expression on CD34⁺ AML blasts compared with autologous CD15⁺ monocytes. (e) HLA class I (W6/32 mAb) expression of CD34⁺ AML blasts ($n = 5$) and CD34⁺CD38⁻ hematopoietic stem cells ($n = 5$) derived from healthy donors. (f) HLA-DR (L243 mAb) expression of CD34⁺ AML blasts ($n = 5$) and CD34⁺CD38⁻ hematopoietic stem cells ($n = 5$) derived from healthy donors. Abbreviations: UPN, uniform patient number.

Comparative profiling of HLA class I ligandomes reveals a multitude of AML-associated antigens

To identify novel targets for peptide vaccination in AML, we comparatively mapped the HLA ligand source proteins of the AML, PBMC and BMNC cohorts. Overlap analysis of HLA source proteins revealed 1435 proteins (23.6% of the mapped AML source proteome) to be exclusively represented in the HLA ligandomes of AML patients. AML was found to share 75.5% (4588 proteins) of its HLA source proteins with PBMCs and 19.3% (1173 proteins) with BMNCs. HLA ligand source proteins of BMNCs showed 89.9%

(1173 proteins) of overlap with the source proteome of PBMCs (Figure 3a). Out of this vast array of potential targets, we aimed to select the most relevant and broadly applicable candidates for off-the-shelf vaccine design. Accordingly, we defined AML exclusivity and high frequency of representation in AML ligandomes as paramount criteria for antigen selection on our platform. Ranking of HLA ligand source proteins according to these criteria identified a subset of 132 proteins (2.2% of the AML source proteome) exclusively represented in $\geq 20\%$ of AML ligandomes (Figure 3b). Within these LiTAAs (ligandome-derived

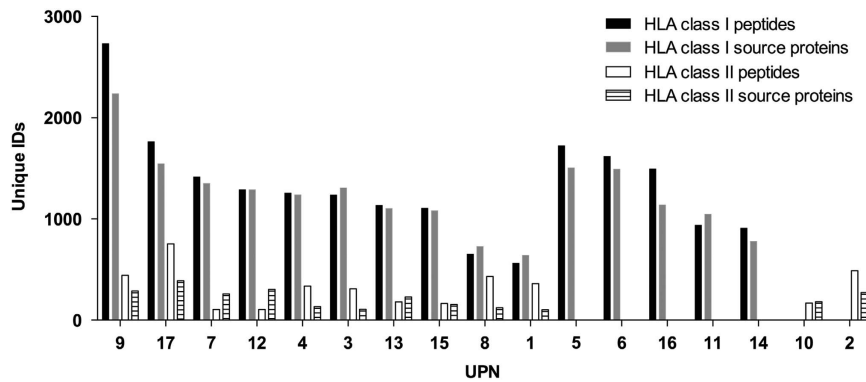


Figure 2. Number of HLA ligand and source protein identifications from primary AML samples. Unique IDs (peptide sequences and corresponding source proteins) identified by LC-MS/MS for HLA class I (W6/32 mAb, $n = 15$) and HLA class II (Tü39 mAb, $n = 12$) in primary AML samples. Only samples fulfilling the threshold of ≥ 500 (HLA class I) and ≥ 100 (HLA class II) unique ligand identifications per sample were included in this study. ID, identification; UPN, uniform patient number.

tumor-associated antigens) defined by these two criteria we identified FAF1 (FAS associated factor 1) as highest ranking, which was detected in 8/15 (53.3%) of patient ligandomes and was represented by six different HLA ligands (AEQFRLEQI (B*44), FTAEFSSRY (A*03), HHDESVLTVNF (B*38:01), REQDEAYRL (B*44:25), RPVMPSRQI (B*07), VQREYNLNF (B*15)). Strikingly, some of these peptides are presented in up to 100% of patients with the appropriate HLA allotype. An overview of the top 15 LiTAAs that showed representation in $\geq 33\%$ of AML ligandomes and their corresponding HLA ligands (LiTAPs) is shown in Table 2 (spectral data is provided in Supplementary Figures S2–S51). In summary, the top 132 most frequent LiTAAs alone provide a panel of 341 different LiTAPs of more than 25 different HLA restrictions, suited for the development of broadly applicable AML-specific peptide vaccines (Supplementary Table S3). The HLA annotation of HLA class I ligands was performed using *in silico* prediction.

In addition, the further 1389 AML-exclusive source proteins with representation frequencies $< 20\%$ represented by 1727 different HLA ligands may serve as repositories for more individualized vaccine design approaches.

Identification of naturally presented HLA class I ligands derived from established AML-associated antigens

A secondary data mining approach focused on the identification and ranking of established AML-associated antigens (as summarized in Anguille *et al.*¹⁵) in our data set of naturally presented HLA ligands. We were able to identify 122 different HLA ligands representing 29 of these published antigens (Supplementary Table S4). In accordance with the findings of a previous study analyzing putative AML antigens on the RNA transcript level,¹⁶ we found $> 80\%$ (24/29) of these antigens also to be represented on benign PBMCs and/or BMNCs, and thus not to be AML-specific. As *WT1* has been described to be highly overexpressed in 70% of AMLs,¹⁶ we assessed the correlation of *WT1* expression on RNA transcript level and HLA-restricted presentation of *WT1*-derived peptides in four AML samples. *WT1* mRNA was detectable in all four samples, and no connection of relative mRNA expression levels and HLA-restricted *WT1* peptide presentation became apparent (Supplementary Table S5). Of note, we found five putative AML antigens exclusively represented on normal PBMCs (TERT, MUC1, SSX2IP, DNAJC2 and PRAME). AML exclusivity with regard to HLA presentation was found for FMS-like tyrosine kinase 3 (FLT3) (SELKMMTQL, B*40), PASD1 (LLGHLPAEI, C*01:02), HOXA9 (DAADELVSQRY, A*26:01), AURKA (REVEIQSHL, B*49:01) and CCNA1 (LEADPFLKY, B*18:01; EPPAVLLL, B*51:01). For myeloperoxidase, we identified a total of 19 different HLA ligands and

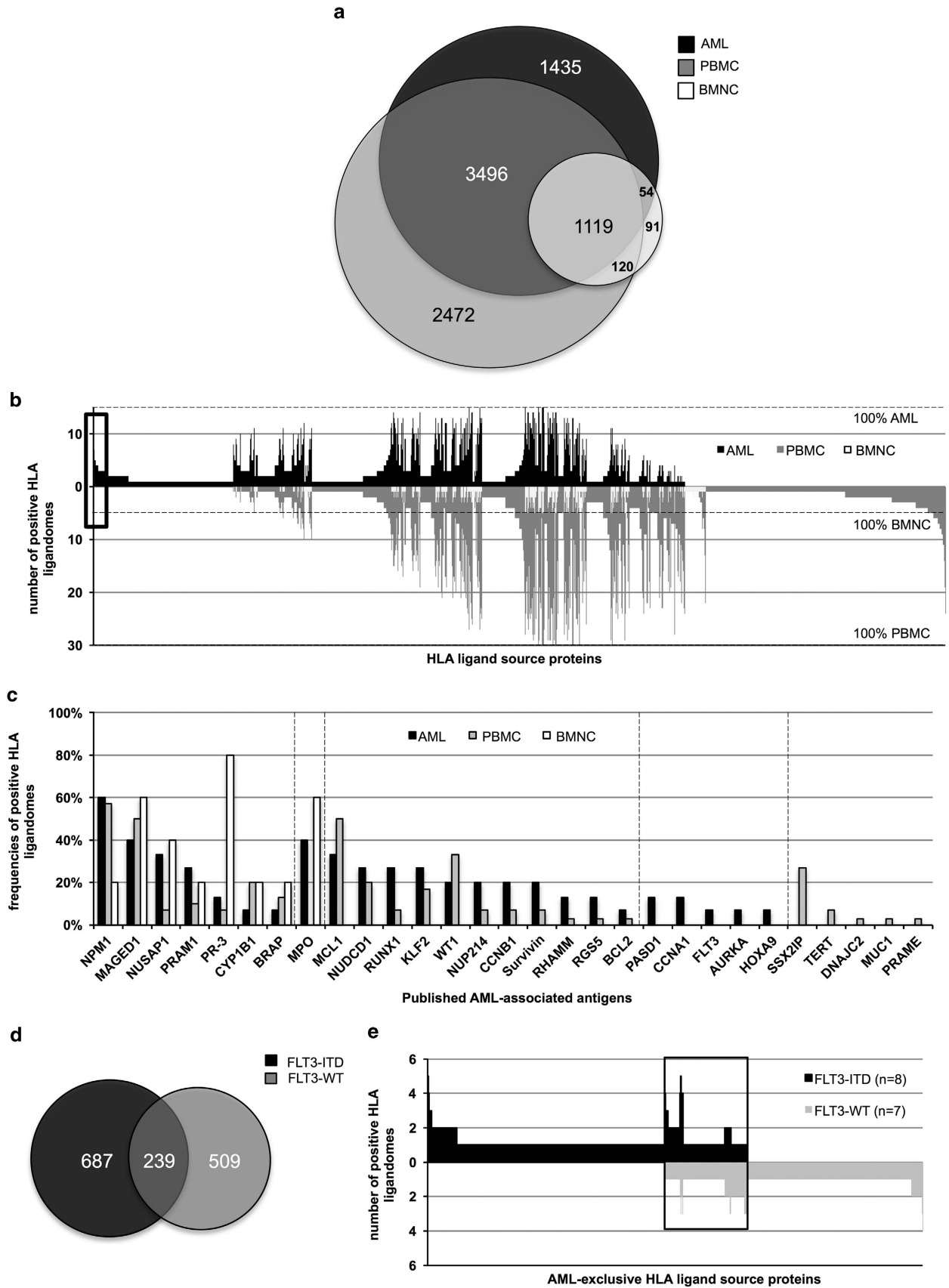
detected representation in 6/15 (40%) of AMLs and 0/30 of PBMC ligandomes. However, analysis of normal BMNCs revealed representation in 3/5 (60%) of ligandomes underlining the relevance of using both PBMCs and BMNCs for target identification. In summary, our analysis underscores the recently proposed concept that a large proportion of previously proposed AML antigens lack AML association¹⁶ by providing additional evidence directly on the HLA ligandome level.

Subset-specific analysis of FLT3-ITD-mutated versus FLT3-WT AML HLA class I ligandomes identifies shared LiTAAs in spite of significant ligandome dissimilarity

To assess the applicability of our novel targets across different subsets of AML, we characterized representation of LiTAAs in FLT3-internal tandem duplication (FLT3-ITD, $n = 8$) and FLT3-wild-type (FLT3-WT, $n = 7$) patient subsets. Similarity indexing of HLA class I ligandomes revealed that the FLT3-WT subset displayed significantly lower internal heterogeneity (mean 916.2 ± 70.6 , $n = 21$) than the FLT3-ITD subset (mean 1687.0 ± 156.5 , $n = 21$, $P < 0.0001$; Supplementary Figure S52). Overlap analysis of AML-exclusive HLA source proteins (FLT3-WT: 748 proteins; FLT3-ITD: 926 proteins) revealed overlaps of 32.0% (FLT3-WT/FLT3-ITD) and 25.6% (FLT3-ITD/FLT3-WT), respectively (Figure 3d). Of note, 42/46 (91.3%) high-ranking LiTAAs were found to be represented in both subsets (Figure 3e). The three HLA ligand source proteins SKP1 (5/8), C16orf13 (5/8) and ERLIN1 (4/8), which were identified exclusively in the FLT3-ITD subset, reached representation frequencies of $\geq 50\%$. The FLT3-WT-specific LiTAA MUL1 was represented in 4/7 (57.1%) of FLT3-WT ligandomes. Taken together, these data support the devised strategy of cohort-comprising analysis of HLA ligandomes for target selection while pointing out a small fraction of highly frequent, subset-specific targets.

Functional characterization of LiTAPs reveals AML-associated immunoreactivity

To evaluate the immunogenicity and specificity of our HLA-A*03 LiTAPs, we next performed 12-day recall IFN- γ ELISPOT assays. PBMCs obtained from six AML patients as well as eight healthy individuals were stimulated with different pools (P_1 and P_2) of top-ranking HLA-A*03 LiTAPs. Significant IFN- γ secretion was observed for both, P_1 and P_2 in 2/6 AML samples (Figure 4a). To confirm these findings, intracellular cytokine staining and flow cytometry for IFN- γ and TNF- α was carried out using 12-day prestimulated PBMCs (Figure 4b). We confirmed P_1 - and P_2 -specific CD8⁺ T-cell responses functionally characterized by IFN- γ (P_1 : 1.6% and P_2 : 1.7% of CD8⁺ T cells) and TNF- α (P_1 : 2.6%



and P_2^I : 2.4% of $CD8^+$ T cells) secretion. Cross-checking ELISPOT assays using A*03-positive healthy donor PBMCs stimulated with P_1^I and P_2^I showed no significant secretion of IFN- γ (0/8; Figure 4c). These initial characterizations demonstrate the here defined LiTAPs to function potentially as AML-specific T-cell epitopes.

HLA class II ligandome analysis provides additional targets and pinpoints potentially synergistic embedded ligands

Overlap analysis of HLA class II source proteomes identified 396 proteins (44.7%) represented by 1079 different HLA ligands to be exclusively represented in the HLA ligandome of AML. AML was found to share 53.3% (472 proteins) and 15.1% (134 proteins) of its source proteome with PBMCs and BMNCs, respectively. BMNCs showed 88.2% (127 proteins) source proteome overlap with PBMCs (Figure 5a). Performing comparative HLA source proteome profiling (Figure 5b) as described above for HLA class I, we were able to identify 36 LiTAAs (represented by 152 different HLA class II ligands) with representation frequencies $\geq 20\%$ (Supplementary Table S6). The highest-ranking class II LiTAA (A1BG) was identified on 6/12 (50%) patients represented by five different ligands. Top-ranking LiTAAs with representation frequencies of $>33\%$ are shown in Table 3 (spectral data provided in Supplementary Figure S53–110). To identify LiTAAs presented in both, the HLA class I and class II ligandomes, we subsequently compared the respective AML-exclusive source proteomes (Figure 5c). This revealed a panel of only 43 shared source proteins (3.0%/10.4% of the HLA class I/HLA class II source proteome, respectively). Mapping of the respective class I against class II ligands identified three HLA class II peptides containing complete embedded HLA class I ligands (Figure 5d).

To characterize functionally the top-ranking HLA class II LiTAPs, we performed 12-day recall IFN- γ ELISPOT assays. For 3/7 top-ranking LiTAPs, significant secretion of IFN- γ was detected in AML patients. T-cell responses were detected for the peptide P_1^I (CLSTN1_{836–852}) in 4/15 (26.7%), for P_2^I (LAB5A_{123–138}) in 3/15 (20.0%) and for P_3^I (MBL2_{191–206}) in 2/15 (13.3%) of AML patients. Figure 5e shows an example of the frequency of specific T cells for peptides P_1^I , P_2^I and P_3^I in an AML patient. Cross-checking ELISPOT assays using healthy donor PBMCs stimulated with P_1^I , P_2^I and P_3^I showed no significant secretion of IFN- γ (0/8). Thus, the here defined AML-specific HLA class II epitopes have the potential to complement an HLA class I peptide vaccine.

DISCUSSION

For a long time peptide-based immunotherapy fell short of its potential to achieve responses both *in vitro* and in the clinical setting of cancer therapy.³⁵ A striking disequilibrium between identified tumor-associated targets on the input side and functional vaccine candidates on the output side became apparent. This might in part have been due to the prevalent approach gene expression profiling for target selection, which does not take into account the distorted relationship of gene

expression and HLA-restricted presentation of the corresponding gene product.^{25,26} A recent study demonstrates that vaccination with tumor-associated peptides identified by direct analysis of primary tumor-derived HLA ligands is able to elicit specific vaccine-induced immune responses that are associated with improved clinical outcome.⁶ This underscores the importance and potential of selecting physiologically relevant targets by direct differential analysis of the entirety of HLA-presented peptides, termed the HLA ligandome.

In this study, we implemented this approach for the identification of novel, naturally presented AML-associated self HLA ligands as a basis for developing an off-the-shelf multi-peptide vaccine for immunotherapy of AML. The prerequisite for peptide-based immunotherapy to be effective is the expression of HLA molecules on target cells. We were able to verify quantitatively sufficient surface expression of HLA class I and II on primary AML samples of five patients. No loss or downregulation of surface HLA was detected on AML blasts compared with autologous normal leukocytes. Similar findings have been reported in previous studies.^{14,36} Mean expression levels of HLA class I of $>100\,000$ and HLA class II of $>17\,000$ do validate AML cells as potential T-cell targets.

Analysis of HLA class I ligand extracts of 15 AML, 30 PBMC and 5 BMNC samples by LC-MS/MS enabled us to comprehensively and comparatively map the HLA ligandomes of the respective cohorts and lead to the identification of a total of more than 25 000 different HLA ligands. Here we devised a novel strategy for target prioritization based on AML exclusivity and frequency of representation, specifically tailored to the identification of broadly applicable vaccine candidates. For this goal, we specifically focused on non-mutant AML-associated self-peptides. Of note, this approach is not able to identify AML-specific, somatic mutation-derived neoantigens,³⁷ which may also represent excellent targets.

Out of the plethora of the identified 1435 AML-exclusive source proteins, our platform highlighted a subset of 132 highly frequent ($>20\%$) LiTAAs and the 341 corresponding LiTAPs. Previous studies in AML based on single-peptide vaccination showed promising results in terms of *in vivo* immunogenicity as well as clinical responses in single patients.^{1,4,21,22} However, the clinical reality of patient and tumor individuality might hamper the broad applicability of non-personalized single-peptide vaccines. Our approach aims at the combination of several peptides derived from different highly frequent LiTAAs in a multi-peptide vaccine, to statistically allow for a broad coverage of the AML patient population. Performing subset-specific HLA ligandome profiling of FLT3-ITD AML as the most prevalent mutated subset,³⁸ we could demonstrate that $>90\%$ of the here defined LiTAAs were represented on both FLT3-ITD and FLT3-WT AML, confirming a broad and subset-unspecific coverage.

Effective recognition of tumor-associated peptides by the immune system is a fundamental requirement for an effective T-cell-based immunotherapeutic approach.³⁹ We performed a functional *in vitro* characterization for an HLA-A*03-restricted

Figure 3. Identification of peptide vaccine targets based on the characterization of the HLA class I ligandomes/source proteomes of AML ($n=15$), PBMC ($n=30$) and BMNC ($n=5$) (a) Overlaps of the HLA class I ligand source proteins of AML, PBMC and BMNC. (b) Comparative profiling of HLA class I ligand source proteins based on the frequency of HLA-restricted representation in AMLs, PBMCs and BMNCs. Absolute number of patients/donors positive for HLA-restricted presentation of the respective source protein (x axis) are indicated on the y axis. Dashed lines indicate 100% representation for each respective cohort. The box on the left-hand side highlights the subset of source proteins showing AML-exclusive representation with frequencies $>20\%$ (LiTAAs). (c) Representation analysis of published AML-associated antigens in HLA class I ligandomes. Bars indicate relative representation of respective antigens by HLA class I ligands in AMLs, PBMCs and BMNCs. (d) Subset-specific analysis of FLT3-ITD-mutated ($n=8$) versus FLT3-WT ($n=7$) AML HLA class I ligandomes. Overlap analysis of AML-exclusive source proteins (as defined in (b)) for FLT3-ITD and FLT3-WT AMLs. (e) Comparative profiling of AML-exclusive HLA class I ligand source proteins based on the frequency of HLA-restricted representation in FLT3-ITD and FLT3-WT AMLs. The box in the middle highlights the subset of shared source proteins, which includes 91.3% of the here defined LiTAAs.

Table 2. Highest-ranking HLA class I LiTAAs and LiTAPs

Protein/peptides	Number of positive AMLs (rep. frequency)	Allotype adj. peptide frequency	PSM	HLA
FAF1	8 (53.3%)		41	
AEQFRLEQI	1	1/5	4	B*44
FTAEFSSRY	2	2/5	7	A*03
HHDESVLTNVF	3	3/3	21	B*38:01
REQDEAYRL	1	1/5	1	B*44:25
RPVMPSRQI	1	1/1	1	B*07
VQREYNLNF	1	1/2	7	B*15
PLXND1	7 (46.7%)		25	
GQLPITIQQ	1	1/1	13	B*13:02
RAYADEVAV	1	1/2	2	B*51:01
REDKPPLAV	1	1/2	1	B*49:01
RVKDLDTKY	2	2/2	5	B*15
SEQEMNAHL	1	1/5	2	B*44:25
YVLPLVHSL	1	1/9	2	A*02
LPLRFVWNI	1	1/2	1	B*51:01
GMNN	6 (40.0%)		20	
EVAEHVQYM	3	3/3	14	A*26:01
YMAELIERL	3	3/9	6	A*02
CPQ	6 (40.0%)		28	
ALASLIRSV	5	5/9	27	A*02
TVAEITGSKY	1	1/3	1	A*26:01
ATP5L	5 (33.3%)		68	
EIGKRGIGY	4	4/7	67	A*26/A*03:01
NLVEKTPAL	1	1/9	1	A*02
ITGA5	5 (33.3%)		67	
IEDKAQILL	3	3/5	55	B*49:01/B*40
SIYDDSYLGY	1	1/3	11	A*26:01
TTNHPIPNK		1/1	1	A*11
SKP1	5 (33.3%)		8	
NAAILKKV	2	2/2	2	B*51:01
NYLDIKGLL	1	1/1	1	A*24
YLDIKGLLDV	2	2/9	5	A*02
CHD1L	5 (33.3%)		70	
EEVGDFIQRY	1	1/5	2	B*44:03
EVGDFIQRY	4	4/7	67	A*26:01/A*03
MKDLSLGGVL	1	NA	1	NA
TGFBRAP1	5 (33.3%)		18	
DEFITVHSM	1	1/1	5	B*18:01
EFITVHSM	2	2/2	6	A*23:01
QLDVRLEI	1	1/1	1	B*13:02
TQYIIHNY	1	1/2	6	B*15
NGLY1	5 (33.3%)		39	
EVVDVTWRY	4	4/7	37	A*26/A*03:01
KEALLRDTI	1	1/2	2	B*49:01
APLP2	5 (33.3%)		94	
HGYENPTYK	4	4/5	92	A*03
SLLYKVPYV	1	1/9	2	A*02
KIF2C	5 (33.3%)		52	
AEIPLRMV	1	1/2	14	B*49:01
EVVYRFTAR	1	1/2	2	A*66
FPGLAIKI	2	2/2	10	B*51:01
IYNGKLFDL	1	1/1	1	A*24
IYNGKLFDLL	1	1/1	7	A*24
KEIDVISI	1	1/2	2	B*49:01
LEEQASRQI	1	1/2	13	B*49:01
TRMSTVSEL	1	1/1	3	B*39:01

Table 2. (Continued)

Protein/peptides	Number of positive AMLs (rep. frequency)	Allotype adj. peptide frequency	PSM	HLA
ELP3	5 (33.3%)		6	
SEETFRFEL	1	1/3	4	B*40
KLYPTLVIR	4	4/5	2	A*03
DGKZ	5 (33.3%)		48	
ALRNQATMVQK	2	2/5	1	A*03:01
LLDHAPPEI	3	3/9	47	A*02
MTCH2	5 (33.3%)		29	
GVLGTVVHKG	4	4/5	11	A*03:01
VQFIGRESKY	1	1/2	18	B*15

Abbreviations: Adj., adjusted; APLP2, amyloid beta (A4) precursor-like protein 2; ATP5L, ATP synthase, H⁺ transporting, mitochondrial Fo complex, subunit; CHD1L, chromodomain helicase DNA-binding protein 1 like; DGKZ, diacylglycerol kinase, zeta; ELP3, elongator acetyltransferase complex subunit 3; GCPQ, carboxypeptidase Q; FAF1, Fas (TNFRSF6)-associated factor 1; GMNN, geminin, DNA replication inhibitor; ITGA5, integrin, alpha 5; KIF2C, kinesin family member 2C; LiTAAs, ligandome-derived tumor-associated antigens; LiTAPs, ligandome-derived tumor-associated peptides; MTCH2, mitochondrial carrier 2; NA, not assigned; NGLY1, N-glycanase 1; PLXND1, plexin D1; PSM, peptide spectrum matches; rep., representation; SKP1, S-phase kinase-associated protein 1; TGFBRAP1, transforming growth factor, beta receptor-associated protein 1. Top 15 highest-ranking HLA class I LiTAAs with representation frequencies > 33% in AML patients ($n = 15$) and representing HLA ligands (LiTAPs) annotated with respective HLA restriction as defined by *in silico* motif mapping using SYFPEITHI or an extended in-house database. The relative frequency of LiTAPs being presented in ligandomes of the respective appropriate HLA allotype can be found in column 3.

subset of our LiTAPs and observed specific CD8⁺ T-cell responses in 33% of AML patients but not in healthy controls. These data provide evidence for the prevalence of functional memory T cells specific for LiTAPs exclusively in AML patients, which, albeit low in frequency, might be recruited to mount an effective anticancer response upon peptide vaccination. Furthermore, the induction of previously absent LiTAP-specific immune response might yield clinical benefit. Importantly, these findings underscore the potential of our target identification strategy to pinpoint physiologically relevant, broadly applicable vaccine target candidates.

Drawing on the extensive amount of data revealed by our approach, we determined the actual representation of published AML-associated antigens¹⁵ in the HLA ligandomes of primary AML, PBMC and BMNC samples *ex vivo*. To our knowledge, our study for the first time identifies/validates naturally presented HLA ligands from established AML-associated antigens by mass spectrometry. Among them were AML-exclusive HLA ligands representing PASD1,⁴⁰ FLT3,⁴¹ HOXA9,⁴² AURAK,⁴³ CCNA1.⁴⁴ It should be noted that a majority (80%) of those published antigens were also identified on samples of healthy donors, and thus not fulfilling our main selection criterion of AML exclusivity.

CD4⁺ T cells have been shown to have an important role in the induction and maintenance of CD8⁺ T-cell responses.^{45–48} Furthermore, multiple direct anticancer activities of CD4⁺ T cells have been described.^{49–52} Therefore, an AML-specific multi-peptide vaccine ideally should be complemented by inclusion of additional HLA class II epitopes. Using our platform, we were able to identify 396 AML-exclusive source proteins, highlighting a panel of 36 LiTAAs represented by 152 HLA class II LiTAPs. Immunologic characterization revealed specific CD4⁺ T-cell

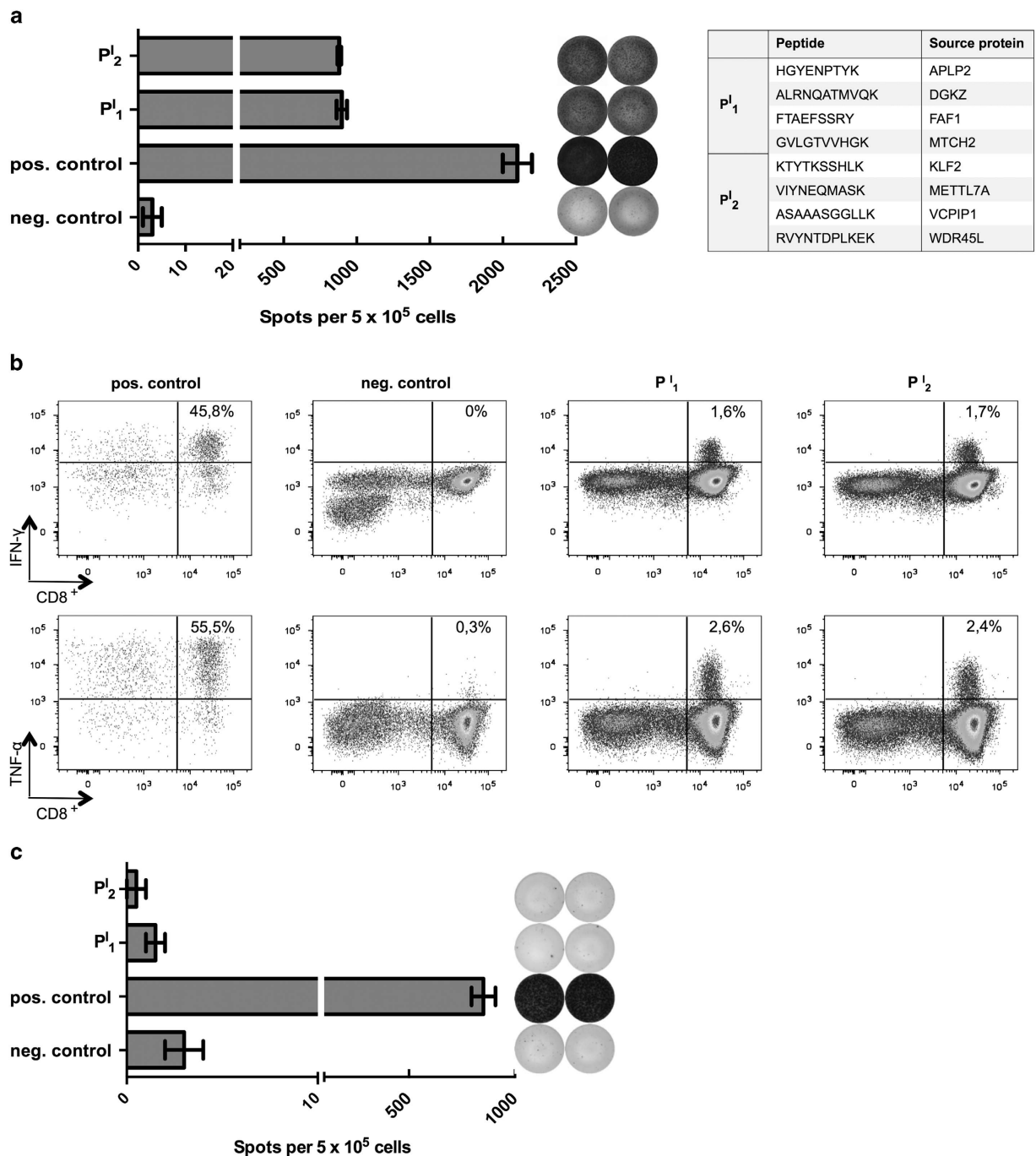


Figure 4. Functional characterization of HLA class I AML-LiTAPs. **(a)** IFN- γ ELISPOT assay of AML patient PBMCs after stimulation with two different A*03-restricted AML LiTAP pools (P₁^I and P₂^I). PHA served as a positive control, and stimulation with HIVGAG_{18–26} A*03 peptide as a negative control. For P₁^I and P₂^I a significant IFN- γ production was observed. **(b)** Intracellular staining for IFN- γ and TNF- α of P₁^I- and P₂^I-stimulated AML patient PBMCs (same as in **(a)**). Phorbol myristate acetate (PMA)/ionomycin served as a positive control, and HIV GAG_{18–26} A*03 peptide as a negative control. **(c)** Cross-checking IFN- γ ELISPOT assay of healthy donor PBMCs after stimulation with A*03-restricted AML LiTAP pools P₁^I and P₂^I revealed no significant IFN- γ production.

responses for 3/7 tested LiTAPs in 13.3%, 20.0% and 26.7% of AML patients, respectively. Notably, three different class II ligands were identified to contain complete, embedded HLA class I peptides, which points to a striking new option for synergistic vaccine design as discussed in recent publications.^{53,54} These novel, naturally presented embedded HLA ligands might present optimal vaccine candidates that are recognized by both, CD4⁺ and CD8⁺ T cells.

In conclusion, this study for the first time comprehensively maps the HLA ligandomes of primary AML samples by direct identification of naturally presented HLA ligands. Devising a novel target selection platform allowed us to pinpoint large panels of AML-exclusive HLA class I and II LiTAAs. The corresponding LiTAPs showed AML-associated immune response patterns *in vitro*, which underscores the potential of our strategy and validates these peptides as prime candidates for novel AML-specific peptide vaccines.

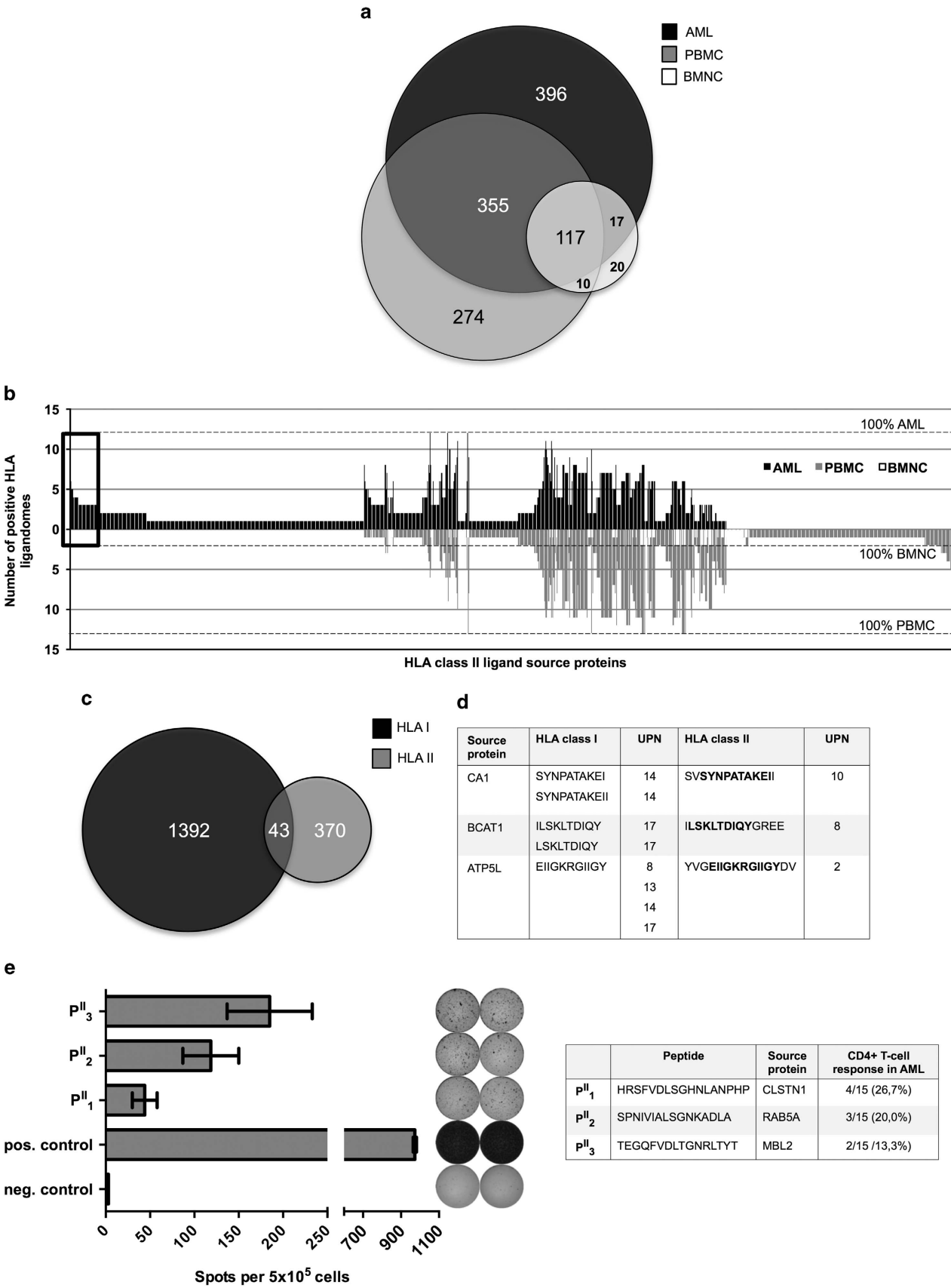


Figure 5. Identification of additional/synergistic peptide vaccine targets based on the characterization of the AML HLA class II ligandome. (a) Overlap analysis of the HLA class II ligand source proteomes of AML ($n = 12$), PBMC ($n = 13$) and BMNC ($n = 2$). (b) Comparative profiling of HLA class II ligand source proteins based on the frequency of HLA-restricted representation in AMLs, PBMCs and BMNCs. Absolute number of patients/donors positive for HLA-restricted presentation of the respective source protein (x axis) are indicated on the y axis. Dashed lines indicate 100% representation for each respective cohort. The box on the left-hand side highlights the subset of source proteins showing AML-exclusive representation with frequencies $> 20\%$. (c) Overlap analysis of HLA class I and HLA class II AML-exclusive source proteins. (d) Of the 43 shared AML-exclusive source proteins, a subset of 3 was identified to contain complete HLA class I ligands embedded in HLA class II peptides. The embedded sequences are depicted in bold. (e) IFN- γ ELISPOT assay of AML patient PBMCs after stimulation with different HLA class II AML LiTAPs (P_{11}^I , P_{12}^I and P_{13}^I). PHA served as a positive control, and stimulation with FLNA_{1669–1683} HLA-DR peptide as a negative control. For P_{11}^I , P_{12}^I and P_{13}^I , a significant increase in IFN- γ production was observed in multiple patients. UPN, uniform patient number.

Table 3. Highest-ranking HLA class II LiTAAs and LiTAPs

Protein/peptides	Number of positive AMLs (rep. frequency)	PSM
A1BG	6 (50.0%)	11
APVELILSDETLPAPE	3	3
ETPDFQLFKNGVAQEPV	1	2
LAPLEGARFALVRED	2	2
SPDRIFHLNAVALGD	1	1
SPDRIFHLNAVALGDG	2	3
CORO1A	5 (41.7%)	11
EEMRKQATVQELQKR	1	1
EEPLSLQELDTSSG	4	5
EMRKQATVQELQKR	1	1
HLEELSLQELDTSSG	1	1
LEEPLSLQELDTSSG	2	3
RPS5	5 (41.7%)	14
AGTVRRQAVDVSPRLR	5	6
IGRAGTVRRQAVDVSPRLR	1	1
RAGTVRRQAVDVSPRLR	4	5
TVRRQAVDVSPRLR	1	2
C19orf10	5 (41.7%)	18
GVVHSFSHNVGPGDK	2	3
GVVHSFSHNVGPGDKY	1	2
GVVHSFSHNVGPGDKYT	1	2
KTEEFVTKTAVAHRP	1	2
KTEEFVTKTAVAHRPG	3	5
VRPGGVVHSFSHNVGPGDK	1	2
VRPGGVVHSFSHNVGPGDKYT	1	2
PLIN3	4 (33.3%)	8
AEKGVRTLAAVSGAQ	1	1
AQPILSKLEPQIASASE	1	1
EKGVRTLTAAVSGAQ	3	3
EKGVRTLAAVSGAQ	1	1
GVRTLAAVSGAQ	1	1
KGVRTLAAVSGAQ	1	1
CLSTN1	4 (33.3%)	8
DVNEYAPVFKEKSYK	1	2
HRSFVDSLGHNLA	1	1
HRSFVDSLGHNLANPH	1	1
HRSFVDSLGHNLANPH	3	4
HSP90B1	4 (33.3%)	19
ALPEFDGKRQFQNAKEG	1	1
DSNEFSVIADPRG	1	2
DSNEFSVIADPRGN	1	2
DSNEFSVIADPRGNT	1	2
DSNEFSVIADPRGNTL	1	2
DSNEFSVIADPRGNTLG	1	2
PEFDGKRQFQNAKE	1	1
SDSNEFSVIADPRGNTLG	1	2
SQKKTFEINPRHLIR	1	1
ALPEFDGKRQFQNAKEG	1	1
PEFDGKRQFQNAKE	2	2
PEFDGKRQFQNAKE	1	1

Table 3. (Continued)

Protein/peptides	Number of positive AMLs (rep. frequency)	PSM
B4GALT1	4 (33.3%)	11
LNSLTYQVLDVQRYYP	1	1
LPQLVGVSSTPLQG	2	3
LPQLVGVSSTPLQGG	1	1
LPQLVGVSSTPLQGGG	3	4
RLPQLVGVSSTPLQGGG	1	1
SDVDLIPmNDHNAYR	1	1
SPN	4 (33.3%)	10
GRRKSROGSLAMEELK	1	2
RKSROGSLAMEELK	3	6
SGPSLKGEEEPVASEDGAVD	1	1
SGSGPSLKGEEEPVASEDGAVD	1	1
METAP1	4 (33.3%)	9
IKPGVTTEEIDHAVH	4	5
KPGVTTEEIDHAVH	3	4
HSPG2	4 (33.3%)	11
DGVLRIQNLDQS	4	5
GAYFHDDGFLAFPG	1	2
TPYSFLPLPTIKDAYR	1	2
YPTPDISWSKLDGSLPP	1	1
YPTPDISWSKLDGSLPPD	1	1

Abbreviations: A1BG, alpha-1-B glycoprotein; AML, acute myeloid leukemia; CLSTN1, calsyntenin 1; CORO1A, coronin, actin-binding protein, 1 A; C19orf10, chromosome 19 open reading frame 10; HLA, human leukocyte antigen; HSP90B1, heat-shock protein 90 kDa beta (Grp94), member 1; HSPG2, heparan sulfate proteoglycan 2; LiTAAs, ligandome-derived tumor-associated antigens; LiTAPs, ligandome-derived tumor-associated peptides; METAP1, methionyl aminopeptidase 1; PLIN3, perilipin 3; PSM, peptide spectrum matches; rep., representation; RPS5, ribosomal protein S5; SPN, sialophorin. Top 11 highest-ranking HLA class II LiTAAs with representation frequencies $> 33\%$ in AML patients ($n = 12$) and representing HLA ligands (LiTAPs).

CONFLICT OF INTEREST

The authors declare no conflict of interest.

ACKNOWLEDGEMENTS

We thank Patricia Hrstic, Beate Pömmel, Claudia Falkenburger, Katharina Graf and Nicole Zuschke for excellent technical support and Mathias Walzer for expert programming of in-house analytical scripts. This work was supported by the Deutsche Forschungsgesellschaft (DFG, SFB 685), the Fortüne Program of the University of Tübingen (Fortüne Number 2032-0-0) and the German Cancer Consortium (DKTK).

REFERENCES

- Mailander V, Scheibenbogen C, Thiel E, Letsch A, Blau IW, Keilholz U. Complete remission in a patient with recurrent acute myeloid leukemia induced by

- vaccination with WT1 peptide in the absence of hematological or renal toxicity. *Leukemia* 2004; **18**: 165–166.
- 2 Feyerabend S, Stevanovic S, Gouttefangeas C, Wernet D, Hennenlotter J, Bedke J et al. Novel multi-peptide vaccination in HLA-A2+ hormone sensitive patients with biochemical relapse of prostate cancer. *Prostate* 2009; **69**: 917–927.
 - 3 Schmitt M, Schmitt A, Rojewski MT, Chen J, Giannopoulos K, Fei F et al. RHAMM-R3 peptide vaccination in patients with acute myeloid leukemia, myelodysplastic syndrome, and multiple myeloma elicits immunologic and clinical responses. *Blood* 2008; **111**: 1357–1365.
 - 4 Oka Y, Tsuboi A, Taguchi T, Osaki T, Kyo T, Nakajima H et al. Induction of WT1 (Wilms' tumor gene)-specific cytotoxic T lymphocytes by WT1 peptide vaccine and the resultant cancer regression. *Proc Natl Acad Sci USA* 2004; **101**: 13885–13890.
 - 5 Rezvani K, Yong AS, Mielke S, Savani BN, Musse L, Superata J et al. Leukemia-associated antigen-specific T-cell responses following combined PR1 and WT1 peptide vaccination in patients with myeloid malignancies. *Blood* 2008; **111**: 236–242.
 - 6 Walter S, Weinschenk T, Stenzl A, Zdrojowy R, Pluzanska A, Szczylak C et al. Multi-peptide immune response to cancer vaccine IMA901 after single-dose cyclophosphamide associates with longer patient survival. *Nat Med* 2012; **18**: 1254–1261.
 - 7 Tallman MS, Gilliland DG, Rowe JM. Drug therapy for acute myeloid leukemia. *Blood* 2005; **106**: 1154–1163.
 - 8 Lowenberg B, Downing JR, Burnett A. Acute myeloid leukemia. *N Engl J Med* 1999; **341**: 1051–1062.
 - 9 Estey E, Dohner H. Acute myeloid leukaemia. *Lancet* 2006; **368**: 1894–1907.
 - 10 Ravandi F, Burnett AK, Agura ED, Kantarjian HM. Progress in the treatment of acute myeloid leukemia. *Cancer* 2007; **110**: 1900–1910.
 - 11 Weiden PL, Flournoy N, Thomas ED, Prentice R, Fefer A, Buckner CD et al. Antileukemic effect of graft-versus-host disease in human recipients of allogeneic marrow grafts. *N Engl J Med* 1979; **300**: 1068–1073.
 - 12 Bleakley M, Riddell SR. Exploiting T cells specific for human minor histocompatibility antigens for therapy of leukemia. *Immunol Cell Biol* 2011; **89**: 396–407.
 - 13 Brouwer RE, van der Heiden P, Schreuder GM, Mulder A, Datema G, Anholts JD et al. Loss or downregulation of HLA class I expression at the allelic level in acute leukemia is infrequent but functionally relevant, and can be restored by interferon. *Hum Immunol* 2002; **63**: 200–210.
 - 14 Vollmer M, Li L, Schmitt A, Greiner J, Reinhardt P, Ringhoffer M et al. Expression of human leucocyte antigens and co-stimulatory molecules on blasts of patients with acute myeloid leukaemia. *Br J Haematol* 2003; **120**: 1000–1008.
 - 15 Anguille S, Van Tendeloo VF, Berneman ZN. Leukemia-associated antigens and their relevance to the immunotherapy of acute myeloid leukemia. *Leukemia* 2012; **26**: 2186–2196.
 - 16 Goswami M, Hensel N, Smith BD, Prince GT, Qin L, Levitsky HI et al. Expression of putative targets of immunotherapy in acute myeloid leukemia and healthy tissues. *Leukemia* 2014; **28**: 1167–1170.
 - 17 Cobbold M, De La Pena H, Norris A, Polefrone JM, Qian J, English AM et al. MHC class I-associated phosphopeptides are the targets of memory-like immunity in leukemia. *Sci Transl Med* 2013; **5**: 203ra125.
 - 18 Van Driessche A, Berneman ZN, Van Tendeloo VF. Active specific immunotherapy targeting the Wilms' tumor protein 1 (WT1) for patients with hematological malignancies and solid tumors: lessons from early clinical trials. *Oncologist* 2012; **17**: 250–259.
 - 19 Li Z, Oka Y, Tsuboi A, Fujiki F, Harada Y, Nakajima H et al. Identification of a WT1 protein-derived peptide, WT1, as a HLA-A 0206-restricted, WT1-specific CTL epitope. *Microbiol Immunol* 2008; **52**: 551–558.
 - 20 Ma Q, Wang C, Jones D, Quintanilla KE, Li D, Wang Y et al. Adoptive transfer of PR1 cytotoxic T lymphocytes associated with reduced leukemia burden in a mouse acute myeloid leukemia xenograft model. *Cytotherapy* 2010; **12**: 1056–1062.
 - 21 Greiner J, Schmitt A, Giannopoulos K, Rojewski MT, Gotz M, Funk I et al. High-dose RHAMM-R3 peptide vaccination for patients with acute myeloid leukemia, myelodysplastic syndrome and multiple myeloma. *Haematologica* 2010; **95**: 1191–1197.
 - 22 Van Tendeloo VF, Van de Velde A, Van Driessche A, Cools N, Anguille S, Ladell K et al. Induction of complete and molecular remissions in acute myeloid leukemia by Wilms' tumor 1 antigen-targeted dendritic cell vaccination. *Proc Natl Acad Sci USA* 2010; **107**: 13824–13829.
 - 23 Qazilbash MH, Rios R, Lu SJ, Giral S. Vaccination with the PR1 leukemia-associated antigen can induce complete remission in patients with myeloid leukemia. *Blood* 2004; **104**: 77A.
 - 24 Qazilbash MH, Thall PF, Wang X, Rios RL, Lu S. PR1 vaccine elicited immunological response after hematopoietic stem cell transplantation is associated with better clinical response and event-free survival. *Blood* 2007; **110**: 178A.
 - 25 Stickel JS, Weinzierl AO, Hillen N, Drews O, Schuler MM, Hennenlotter J et al. HLA ligand profiles of primary renal cell carcinoma maintained in metastases. *Cancer Immunol Immunother* 2009; **58**: 1407–1417.
 - 26 Weinzierl AO, Lemmel C, Schoor O, Muller M, Kruger T, Wernet D et al. Distorted relation between mRNA copy number and corresponding major histocompatibility complex ligand density on the cell surface. *Mol Cell Proteomics* 2007; **6**: 102–113.
 - 27 Neumann A, Horzer H, Hillen N, Klingel K, Schmid-Horch B, Buhring HJ et al. Identification of HLA ligands and T-cell epitopes for immunotherapy of lung cancer. *Cancer Immunol Immunother* 2013; **62**: 1485–1497.
 - 28 Neidert MC, Schoor O, Trautwein C, Trautwein N, Christ L, Melms A et al. Natural HLA class I ligands from glioblastoma: extending the options for immunotherapy. *J Neurooncol* 2013; **111**: 285–294.
 - 29 Falk K, Rotzschke O, Stevanovic S, Jung G, Rammensee HG. Allele-specific motifs revealed by sequencing of self-peptides eluted from MHC molecules. *Nature* 1991; **351**: 290–296.
 - 30 Kowalewski DJ, Stevanovic S. Biochemical large-scale identification of MHC class I ligands. *Methods Mol Biol* 2013; **960**: 145–157.
 - 31 Kall L, Canterbury JD, Weston J, Noble WS, MacCoss MJ. Semi-supervised learning for peptide identification from shotgun proteomics datasets. *Nat Methods* 2007; **4**: 923–925.
 - 32 Sturm T, Leinders-Zufall T, Macek B, Walzer M, Jung S, Pommerl B et al. Mouse urinary peptides provide a molecular basis for genotype discrimination by nasal sensory neurons. *Nat Commun* 2013; **4**: 1616.
 - 33 Widenmeyer M, Griesemann H, Stevanovic S, Feyerabend S, Klein R, Attig S et al. Promiscuous survivin peptide induces robust CD4+ T-cell responses in the majority of vaccinated cancer patients. *Int J Cancer* 2012; **131**: 140–149.
 - 34 Britten CM, Gouttefangeas C, Welters MJ, Pawelec G, Koch S, Ottensmeier C et al. The CIMT-monitoring panel: a two-step approach to harmonize the enumeration of antigen-specific CD8+ T lymphocytes by structural and functional assays. *Cancer Immunol Immunother* 2008; **57**: 289–302.
 - 35 Mellman I, Coukos G, Dranoff G. Cancer immunotherapy comes of age. *Nature* 2011; **480**: 480–489.
 - 36 Wetzler M, Baer MR, Stewart SJ, Donohue K, Ford L, Stewart CC et al. HLA class I antigen cell surface expression is preserved on acute myeloid leukemia blasts at diagnosis and at relapse. *Leukemia* 2001; **15**: 128–133.
 - 37 Cancer Genome Atlas Research Network. Genomic and epigenomic landscapes of adult *de novo* acute myeloid leukemia. *N Engl J Med* 2013; **368**: 2059–2074.
 - 38 Patel JP, Gonen M, Figueroa ME, Fernandez H, Sun Z, Racevskis J et al. Prognostic relevance of integrated genetic profiling in acute myeloid leukemia. *N Engl J Med* 2012; **366**: 1079–1089.
 - 39 Cheever MA, Allison JP, Ferris AS, Finn OJ, Hastings BM, Hecht TT et al. The prioritization of cancer antigens: a national cancer institute pilot project for the acceleration of translational research. *Clin Cancer Res* 2009; **15**: 5323–5337.
 - 40 Hardwick N, Buchan S, Ingram W, Khan G, Vittes G, Rice J et al. An analogue peptide from the Cancer/Testis antigen PASD1 induces CD8+ T cell responses against naturally processed peptide. *Cancer Immunol* 2013; **13**: 16.
 - 41 Graf C, Heidl F, Tenzer S, Radsak MP, Solem FK, Britten CM et al. A neoepitope generated by an FLT3 internal tandem duplication (FLT3-ITD) is recognized by leukemia-reactive autologous CD8+ T cells. *Blood* 2007; **109**: 2985–2988.
 - 42 Beatty GL, Smith JS, Reshef R, Patel KP, Colligon TA, Vance BA et al. Functional unresponsiveness and replicative senescence of myeloid leukemia antigen-specific CD8+ T cells after allogeneic stem cell transplantation. *Clin Cancer Res* 2009; **15**: 4944–4953.
 - 43 Ochi T, Fujiwara H, Suemori K, Azuma T, Yakushiji Y, Hato T et al. Aurora-A kinase: a novel target of cellular immunotherapy for leukemia. *Blood* 2009; **113**: 66–74.
 - 44 Ochsenreither S, Majeti R, Schmitt T, Stirewalt D, Keilholz U, Loeb KR et al. Cyclin-A1 represents a new immunogenic targetable antigen expressed in acute myeloid leukemia stem cells with characteristics of a cancer-testis antigen. *Blood* 2012; **119**: 5492–5501.
 - 45 Greiner J, Ono Y, Hofmann S, Schmitt A, Mehrling E, Gotz M et al. Mutated regions of nucleophosmin 1 elicit both CD4(+) and CD8(+) T-cell responses in patients with acute myeloid leukemia. *Blood* 2012; **120**: 1282–1289.
 - 46 Schoenberger SP, Toes RE, van der Voort EI, Offringa R, Melief CJ. T-cell help for cytotoxic T lymphocytes is mediated by CD40–CD40L interactions. *Nature* 1998; **393**: 480–483.
 - 47 Sun JC, Williams MA, Bevan MJ. CD4+ T cells are required for the maintenance, not programming, of memory CD8+ T cells after acute infection. *Nat Immunol* 2004; **5**: 927–933.
 - 48 Janssen EM, Lemmens EE, Wolfe T, Christen U, von Herrath MG, Schoenberger SP. CD4+ T cells are required for secondary expansion and memory in CD8+ T lymphocytes. *Nature* 2003; **421**: 852–856.

- 49 Mumberg D, Monach PA, Wanderling S, Philip M, Toledano AY, Schreiber RD *et al*. CD4(+) T cells eliminate MHC class II-negative cancer cells in vivo by indirect effects of IFN-gamma. *Proc Natl Acad Sci USA* 1999; **96**: 8633–8638.
- 50 Qin ZH, Blankenstein T. CD4(+) T cell-mediated tumor rejection involves inhibition of angiogenesis that is dependent on IFN gamma receptor expression by nonhematopoietic cells. *Immunity* 2000; **12**: 677–686.
- 51 Muller-Hermelink N, Braumuller H, Pichler B, Wieder T, Mailhammer R, Schaak K *et al*. TNFR1 signaling and IFN-gamma signaling determine whether T cells induce tumor dormancy or promote multistage carcinogenesis. *Cancer Cell* 2008; **13**: 507–518.
- 52 Perez-Diez A, Joncker NT, Choi K, Chan WF, Anderson CC, Lantz O *et al*. CD4 cells can be more efficient at tumor rejection than CD8 cells. *Blood* 2007; **109**: 5346–5354.
- 53 Tomita Y, Yuno A, Tsukamoto H, Senju S, Kuroda Y, Hirayama M *et al*. Identification of promiscuous KIF20A long peptides bearing both CD4+ and CD8+ T-cell epitopes: KIF20A-specific CD4+ T-cell immunity in patients with malignant tumor. *Clin Cancer Res* 2013; **19**: 4508–4520.
- 54 Melief CJ. 'License to kill' reflects joint action of CD4 and CD8 T cells. *Clin Cancer Res* 2013; **19**: 4295–4296.
- 55 Swerdlow SH, Campo E, Harris NL, Jaffe ES, Pileri SA, Stein H *et al*. (eds). *WHO Classification of Tumours of Haematopoietic and Lymphoid Tissues*, 4th edn. IARC Press: Lyon, France, 2008.

Supplementary Information accompanies this paper on the Leukemia website (<http://www.nature.com/leu>)